

6+2=8
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National University of Computer and Emerging Sciences, Lahore Campus



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Instruction/Notes: Plagiarism in quiz from any source (internet or a student) will result in F grade.

Q1: A robot is placed in cell A and its goal is to reach cell Y in the given 5x5 grid. The allowed movement order is (Up, Down, Right, Left). The cells B, R, P, U, W, and H are blocked and cannot be visited.

Assume the following: Step cost: Cost = 1 for nodes at even depth levels, Cost = 2 for nodes at odd depth levels

- Heuristic function $h(n)$: Use Manhattan distance from the current cell to the goal cell Y.

Answer the following:

a. Draw the search tree using Greedy Best-First Search (3)

c. Write the evaluation function used in both searches (2) $f(n) = h(n)$

d. Mention the solution path for search (2)

e. Does Greedy Best-First Search guarantee an optimal and Complete solution? Why or why not? (3)

